

SECTION B – WRITTEN QUESTIONS (Total: 80 marks)

Answer **ALL** questions in this section. Marks are indicated at the end of each question. Together they are worth 80% of the total marks for this examination.

Question 1 (26 marks – approximately 47 minutes)

Holmes Industrial Limited ("HIL") produces and sells a single product. The financial year of HIL ends on 31 December. The following data have been budgeted for the next two years.

	Year 1	Year 2
Production in units	500,000	400,000
Sales in units	360,000	480,000
	HK\$ per unit	
Selling price		200
Direct material cost		80
Direct labour cost		20
Variable production overhead costs		30

There was no opening inventory at the beginning of Year 1. Fixed production overhead costs are HK\$15 per unit, absorbed based on a normal production level of 420,000 units per annum. The variable non-production overhead costs are HK\$10 per unit sold and the annual fixed non-production overhead costs are HK\$5,000,000 for the next two years.

Required:

- (a) Prepare the budgeted statement of profit or loss for Years 1 and 2 for HIL using an absorption costing system.
(10 marks)
- (b) Prepare the budgeted statement of profit or loss for Years 1 and 2 for HIL using a marginal costing system.
(9 marks)
- (c) Prepare the reconciliation of the budgeted profits reported in your answers to Question 1(a) and 1(b) above for each of the next two years.
(4 marks)
- (d) State any **THREE** reasons for using a marginal costing system instead of an absorption costing system for internal decision making.
(3 marks)

Question 2 (26 marks – approximately 47 minutes)

Fontaine Manufacturing Limited ("FML") manufactures bathroom utensils. Its factory in Tseung Kwan O Industrial Estate is producing two types of basin cabinets with the following cost and sales forecasts for next quarter.

	Aluminium	Plywood
Sales demand (units)	12,000	10,000
	HK\$ per unit	HK\$ per unit
Selling price	2,900	4,000
Direct material cost	850	1,400
Direct labour cost	300	400
Variable production overhead	250	600
Fixed production overhead	600	500

FML's production capacity is estimated at 62,000 direct labour hours for next quarter. Direct labour will be paid at HK\$100 per hour.

Required:

- (a) Calculate the shortage in direct labour hours for FML if production is scheduled to meet all the expected sales demands of the two products. (4 marks)
- (b) Determine the production priority of the two products in view of the labour constraint. Prepare an optimal production plan and calculate the total contribution margin of each product and the profit of FML as a whole for next quarter to maximise company profit. (16 marks)
- (c) In view of the shortage in direct labour hours to meet all the sales demands, explain to the management of FML THREE ways to overcome the labour shortage. (6 marks)

Question 3 (18 marks – approximately 32 minutes)

The management accountant of Karina Toys Company ("KaTC") has reported the following data for the past three quarters for one of its products, Emerald.

	1 st Quarter	2 nd Quarter	3 rd Quarter
Output in units	20,000	24,000	22,000
	HK\$	HK\$	HK\$
Material costs	600,000	720,000	660,000
Labour costs	1,200,000	1,400,000	1,300,000
Production overhead costs	<u>1,500,000</u>	<u>1,500,000</u>	<u>1,500,000</u>
Production costs	<u>3,300,000</u>	<u>3,620,000</u>	<u>3,460,000</u>

Required:

- (a) Explain the type of cost behaviour of labour costs. (2 marks)
- (b) Calculate the budgeted production costs with detailed cost elements for the 4th quarter with an expected output of 25,000 units of Emerald using the high-low method. (5 marks)
- (c) KaTC is preparing the budgets for the 1st quarter of next year. The following sales demand information has been predicted for its other product, Sapphire.

<u>January</u>	<u>February</u>	<u>March</u>	<u>April</u>
30,000 units	28,000 units	40,000 units	35,000 units

KaTC will hold a closing inventory of finished goods of Sapphire, which is equal to 40% of the following month's sales demand. 10% of the production output is expected to be faulty with nil scrap value at the end of the production process. Prepare the production budget of Sapphire in units by month for the 1st quarter of next year.

Note: Round the units to the nearest integer.

(11 marks)

Question 4 (10 marks – approximately 18 minutes)

iSports Products Limited ("iSPL") specialises in designing, developing, manufacturing and marketing products including sportswear, footwear and apparel and accessories. You are provided with the following information related to last year.

	HK\$
Total assets	52,000,000
Operating assets	32,000,000
Operating profit	5,760,000
Company required rate of return	15%

Required:

- (a) Calculate the residual income of last year for iSPL and analyse the result.
(4 marks)
- (b) Apart from financial measures, the management of iSPL is considering also using non-financial measures to evaluate its performance. Explain any THREE advantages of using non-financial measures to track company performance.
(6 marks)

* * * END OF EXAMINATION PAPER * * *